Fratricide Risk in an Integrated Air & Missile Defence System of Systems

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Abstract

This paper describes a method for analysing fratricide hazards in an Integrated Air and Missile Defence (IAMD) System of Systems (SoS). Literature and case studies were used to trace causal mechanisms through five distributed functional threads and derive 44 hazard controls. Event-tree and fault-tree analysis supported control allocation and the examination of dependencies. A worked fault tree for airspace protection identified several controls dependent on a common aeronautical information service. The wider review also identified dependencies on IFF, Link 16 and network time distribution. A supplementary case prompted changes to the hazard definition, event tree and dynamic separation controls. The paper also considers how the controls could be assessed and implemented through successive Agile capability increments.

Keywords: Constituent systems, fratricide, system of systems, enterprise system, hazard analysis, risk assessment

1 Introduction

Fratricide is an inherent risk of military operations. Australian Defence Force (ADF) doctrine requires commanders to incorporate preventive controls into planning and execution (ADF 2009b). In an Integrated Air and Missile Defence (IAMD) System of Systems (SoS), these controls depend on functions distributed across independently managed sensors, communications networks, command and control systems and weapon platforms. Sensor netting, remote cueing and Integrated Fire Control (IFC) extend this dependence across the engagement chain (Williams 2026; Grant 2002).

Fratricide hazards can therefore arise from interactions among constituent systems. Stale position reports, incorrect track associations or conflicting airspace constraints may each appear tolerable locally yet combine to expose a friendly aircraft to engagement. The analysis must follow these interactions across the enterprise, including the people and procedures that coordinate its systems (INCOSE 2026; Snook 2011).

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Unfortunately there is no generally accepted single method for SoS Hazard Analysis (SoSHA). Methods have been developed to address interface hazards, behavioural deviations, reconfiguration and network dependencies, but a unified approach remains elusive (Redmond, Michael, and Shebalin 2008; Despotou, Alexander, and Kelly 2009; Aitken, Alexander, and Kelly 2010; Axelsson 2020). Changing system boundaries, independent operating modes and limited access to constituent designs further complicate the SoSHA task.

The immediate need was thus to develop and demonstrate an analysis method sufficient to identify controls for fratricide that could be allocated through the integrative C2 element of the ADF's IAMD SoS as the SoS evolved (Australian DoD 2026).

Refer to Table 5 for terms and definitions.

2 Background and case studies

The literature review combined searches of the US Defense Technical Information Center (DTIC) and Google Scholar for fratricide and blue-on-blue events with relevant US and Australian doctrine. Accident reports and prior analyses were used to identify causal mechanisms, dependencies and controls. The review covered architecture and decision making, autonomy and human coordination, and the effects of closer integration.

2.1 Architecture and decision making

The critical air defence decision is whether to engage a target. This decision is rarely made by the weapon platform alone; instead, it is shaped upstream by rules of engagement, force level weapons control status, delegated authority and airspace control measures. Although ADF doctrine places ultimate engagement authority with the C2 agency holding the best information, the decisions and information supporting engagement are distributed across the SoS and over time (ADF 2009b).

Coordination across an enterprise SoS requires compatible interfaces, a common interpretation of information, agreed procedures and clear authority (Dahmann and Henshaw 2026). ADF command doctrine identifies a common operational picture, supported by common graphics and processes, as a means of coordination and fratricide prevention (ADF 2024). The following examples show how that coordination can fail:

a. A track may be technically valid, but local node clock drifts may result in it being interpreted differently across the system.

b. An Identification Friend or Foe (IFF) return may exist, but is correlated to the wrong radar track, or geographically distributed interrogating systems may correlate it with differing tracks.

c. An airspace control measure may have been authorised appropriately, but has not been received at a local C2 node.

d. A weapon platform may comply with its internal weapons control status while that status remains inconsistent with the global SoS level status.

Where downstream activities assume the correctness of upstream data and planning products, inconsistencies and errors can produce errors in subsequent decisions. Planning errors, such as unresolved conflicts over airspace use, can also remain undetected and latent. Where planning is performed at the constituent system level, failure to distribute local plans to all affected units can also result in hazardous undetected conflicts, as the Tornado ZG710 case illustrates (MoD 2004).

2.2 Integration

Greater integration can increase fratricide risk by tightening the coupling between constituent systems, reducing decision time and allowing errors to propagate rapidly across the SoS (Porter, Squair, and Singh 2006). Enterprise integration can occur physically through airspace use, and functionally, as IFC links distributed sensors, command systems and shooters into a single engagement chain. Both forms increase interdependence and create new failure pathways.

Airspace segregation provides a physical means of separating friendly aircraft from own-force missiles. A Joint Engagement Zone permits shared use and consequently places greater reliance on current position, identity and engagement information (JCS 2010). Track-association errors then have more direct consequences for friendly aircraft (Grant 2002).

IFC distributes detection, tracking and engagement functions among constituent systems (Australian DoD 2022; Young 2005). Remote fire and engagement handover extend these dependencies into the weapon-control loop. The safety problem includes preserving state and authority across interfaces and during changes of control.

2.3 Autonomy and human integration

Increasing autonomy affects both the integration of autonomous weapon systems and the human coordination on which the SoS depends.

Autonomous weapon systems rely on constraints defined before the mission. As conditions change, a system may comply with those constraints while conflicting with SoS requirements. Hobbs and Null (2016, slide 28) describe a drone incident in which a mission change resulted in a lost-link procedure directing the aircraft towards terrain. This illustrates the hazard of obsolete assumptions (McFarland 2022). Engagement controls should therefore check that their operating assumptions remain valid and inhibit or terminate action when they do not.

Humans also perform integration work by reconciling conflicting information, clarifying intent and coordinating systems. Laskey and Laskey (2002) illustrate the limitations of identification logic confronted by ambiguous inputs. Operators may compensate for these limitations, but their contribution can be overlooked. Automation can then remove that contribution while leaving operators to handle the difficult exceptions (Bainbridge 1983). Any change in automation must identify the human functions on which fratricide prevention depends and determine how they will be preserved.

2.4 Case studies

Four incidents were used to inform model development. A fifth supplementary case study was assessed separately (Section 8.2). The case studies are summarised below:

a. USS *Vincennes* (1988). Mixed civil-military airspace and track/IFF confusion contributed to the unintended engagement of a civil aircraft (Fogarty 1988; Gruner 1990).

b. Black Hawk Iraq shoot-down (1994). Information about friendly helicopters was available within the force but did not reach the weapons platforms (GAO 1997; Leveson, Allen, and Storey 2002; Snook 2011).

c. Royal Air Force (RAF) Tornado GR4A ZG710 shoot-down (2003). Conflicting airspace use, autonomous local operation and degraded identification cues contributed to the aircraft being classified as an anti-radiation missile (MoD 2004).

d. United States Navy (USN) F/A-18 shoot-down (2003) - supplementary. Postol (2004) proposed that an adjacent radar's electromagnetic interference generated a false ballistic-missile track. Hawley (2017) describes the transition from standby to ready while automatic engagement remained enabled.

e. USS *Gettysburg* (2024). Degraded services and procedural errors led to returning friendly aircraft being interpreted as missile threats (Hakimzadeh 2025; Altman 2025).

A detailed description of associated SoS hazard causal mechanisms is provided in Table 2.

3 Hazard analysis methodology

The method combined literature and case evidence with top-down and bottom-up hazard analysis (Figure 1). An event tree organised the protective barriers, while fault trees examined mechanisms and dependencies that could defeat them. The method comprised five stages:

a. *Literature review.* The literature and case review identified causal mechanisms, dependencies and controls. Causal mechanisms were identified and classified using the SoS taxonomy of Ericson (2015) (see Figure 2). Causal mechanisms that could fit more than one category were assigned to the category that best described how the hazardous interaction arose (Table 2).

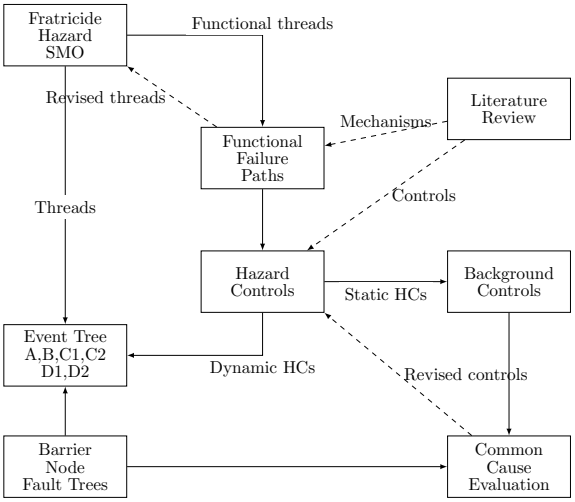


Figure 1: Fratricide modelling methodology

b. *Top-down analysis.* The Source-Mechanism-Outcome (SMO) hazard model of Swallom (2007) was used to define the fratricide hazard. The model was then decomposed into functional threads.

c. *Bottom-up analysis.* The constituent functional threads and their interactions were examined to identify credible functional failure mechanisms that could contribute to a functional failure path. Hazard controls were identified for these causal mechanisms.

d. *Hazard modelling and control allocation.* An event tree was used to organise the hazard controls into time sequenced stages of failure propagation (Figure 3) while supporting fault-tree analysis examined mechanisms and dependencies that could defeat them (Figure 4). Dynamic controls were allocated to barriers and enabling arrangements recorded as background conditions.

e. *Common cause analysis.* For the airspace control thread, the allocated controls were examined to determine whether common cause failures could defeat multiple controls or functional barriers simultaneously.

The analysis was iterative with findings at later stages prompting revision of earlier assumptions and models. The sequence above presents the resulting method in a logical order for the purposes of explanation (Parnas and Clements 1986).

4 Functional and hazard model

The functional model comprises five concurrent, interacting threads that perform the air defence mission and protect friendly platforms:

1. *Tracking.* Maintain a continuous, shared picture of the location and movement of tracked objects.
2. *Friend force tracking.* Maintain a current, shared picture of friendly platform locations and movements.
3. *Identification and classification.* Determine whether tracked objects are friendly, unknown, neutral or hostile, and assess their type, behaviour and inferred intent.

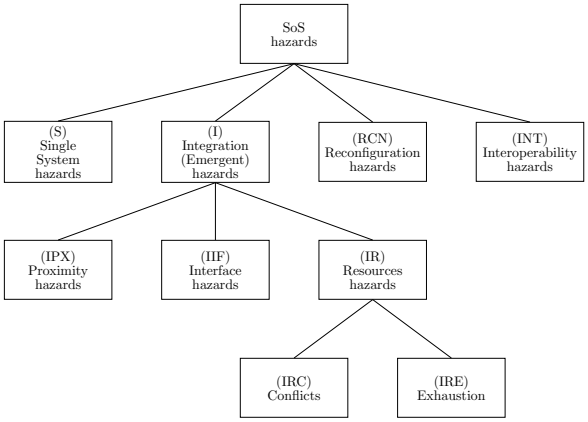


Figure 2: Ericson's SoS hazards taxonomy

4. *Engagement.* Authorise, initiate and control engagements, and stop them while effective intervention remains possible.
5. *Airspace Control.* Manage routes, corridors and defended volumes, including dynamic separation of friendly aircraft from weapon acquisition volumes during engagements.

A supporting aeronautical information service provides a common, current baseline of airfields, routes, procedures and boundaries against which live tracks and proposed engagements can be assessed.

Failures across these interacting functional threads can produce the following SoS fratricide hazard (H1), expressed using the Source-Mechanism-Outcome model of Swallom (2007):

a. *Hazard source.* A friendly platform occupies, or may enter, a region in which a friendly weapon engagement can expose it to harm. This condition can arise through intentional or unintentional sharing of airspace, irrespective of whether the platform is positively identified as friendly.

b. *Causal mechanism.* The SoS fails to maintain or apply the identity, position, airspace and engagement constraints needed to protect the platform. The platform may be selected as a target or acquired during an engagement directed at another contact.

c. *Outcome.* The friendly platform is unintentionally hit or otherwise harmed by a friendly weapon.

This hazard model recognises that fratricide does not require the direct targeting of a friendly platform but can also arise from adjacency to a target being legitimately attacked.

4.1 SoS hazard causal modelling

The SMO model describes the hazard, while the event tree (Figure 3) represents the protective decisions and possible outcomes. Its stages distinguish longer-duration protection from engagement-specific intervention:

a. Barrier A represents sustained airspace protection supported by shared FFT, COP and coordination, typically established over days or hours.

b. Barrier B represents exclusion of a specific friendly contact from the target set through tracking, identification, procedural and engagement logic, typically over hours or minutes.

c. Barriers C1 and C2 represent pre-launch inhibition and effective post-launch intervention, respectively, within the available minutes or seconds.

d. D1 and D2 represent successful aircrew evasion and the weapon failing to hit the friendly platform. These outcomes acknowledge that an unsafe engagement need not result in harm; they are not engineered protective barriers.

Exclusion at B ends the assessed exposure only if the friendly platform is also clear of engagements directed at other contacts. Otherwise assessment continues through event tree barriers C1 and C2. A conflict first arising after launch is assessed at C2, where protection depends on effective intervention in time to prevent harm. The supporting fault trees examine control dependencies.

4.2 Dynamic separation

Dynamic separation protects friendly aircraft from exposure to a missile's seeker acquisition volume and fly-out path. At barrier C1, the control withholds or revokes launch permission where predicted friendly occupancy conflicts with either, or where adequate separation cannot be established. Assessment covers the predicted geometry during fly-out and acquisition, including uncertainty and changes in friendly or threat trajectories. At barrier C2 separation assessment continues and protective action is initiated while effective intervention remains feasible. Applying the control at both barriers does not make the two applications independent.

5 Functional hazard pathways

The five functional threads (Section 4) operate concurrently across the SoS. Each performs mission functions and provides information or constraints needed by other threads to prevent fratricide. The event tree identifies opportunities for intervention within these concurrent functions.

5.1 Tracking

Bath (2002) explained that closely spaced friendly and enemy tracks could swap their streams of detections, potentially making a friendly aircraft engagement eligible. Where track data are distributed as tracklets, message loss, latency and out-of-sequence delivery can increase track uncertainty and contribute to track swaps and misidentification. False contacts provide another pathway to fratricide. For example, Postol (2004) proposed that interference between adjacent radars could generate false tracks. An engagement may therefore be initiated against a represented contact that does not correspond to the physical object subsequently acquired by a weapon system.

These are primarily integration hazards involving correlation, track binding and network transmission, including interface, timing and proximity mechanisms. These mechanisms map primarily to barrier B of Figure 3, while network availability and picture coherence mechanisms map to barrier A. The validity of track information also affects launch inhibition at C1 and recognition of a dangerous engagement at barrier C2.

5.2 Friend force tracking

Fratricide risk rises when a friendly platform is physically present in engagement space but data on its location are absent, stale, mis-registered or distrusted by the decision maker. For example, Bryant and Smith (2013) showed that latency in FFT position reports significantly increased fratricide events, regardless of whether the decision maker was aware of the latency. FFT also supplies the position, movement, currency and uncertainty information needed to assess predicted friendly separation during an engagement. In this context missing, stale or incorrect FFT information can invalidate engagement permissives.

These are primarily integration hazards involving the distribution and use of friendly-position reports. FFT mechanisms affect the shared picture at barrier A of Figure 3. Missing, stale or distrusted reports can also prevent target-set exclusion at B and compromise separation assessment at barriers C1 and C2.

5.3 Identification and classification

Fratricide can occur when ambiguity, uncertainty or conflict in the identification evidence is not presented to the decision maker, causing the identity assessment to appear more certain than the evidence warrants. The USN's Tactical Aid to Decision Making Under Stress (TADMUS) project also found that decision aids should expose supporting and conflicting evidence and the assumptions underlying an identification so that ambiguous data prompts consideration of alternative explanations (Morrison, Kelly, and Hutchins 1996). In contrast, ambiguous or mis-associated evidence supported confident hostile identification in both the Black Hawk and *Vincennes* shoot-downs, allowing the aircraft to remain in the target set.

These are primarily SoS interoperability hazards involving conflicting identification or classification results across systems. Combat Identification (CID) mechanisms map primarily to barrier B of Figure 3, while a final cooperative interrogation at barrier C1 provides another opportunity to detect the error before engagement.

5.4 Engagement

Distributed authority, state and timing can produce an engagement permissive that is locally valid but inconsistent with SoS constraints. Remote fire and forward-pass IFC extend these dependencies into the weapon-control loop (Young 2005). The literature identifies several resulting pathways:

a. Redmond, Michael, and Shebalin (2008) described a SoS reconfiguration hazard in which a BMDS command post handed control of a missile to another command post without transferring its complete state.

b. Hawley (2017) identified an inadvertent launch pathway in the 2003 F/A-18 shoot-down due to mode transition from system standby to ready with the auto-engage function active and an active track eligible for engagement.

c. A safety assessment of the US BMDS by Pereira, Lee, and Howard (2006) identified three related inadvertent launch pathways.

   (a) *Stale track.* An ambiguous definition of an active track allowed a stale track to remain engagement eligible, i.e. an interoperability hazard due to semantic and procedural inconsistency.

   (b) *Mode change.* During a transition from test to live operation, delayed mode feedback could allow a test command to reach a launch station connected to live interceptors, i.e. an reconfiguration hazard due to temporary mode incoherence.

   (c) *Command ordering.* Fire Enable and Fire Disable commands travelled over separate communication paths and could arrive in reverse order because of differing transmission delays, ie. an interface hazard occurs due to an unsafe cross interface race condition.

The Redmond, Michael, and Shebalin (2008) and Pereira, Lee, and Howard (2006) scenarios required no constituent system failure. Each arose from inconsistent state, mode or timing across interacting systems. They also illustrate failure propagation through the event tree of Figure 3; stale track information could defeat target set exclusion (barrier B), mode incoherence could bypass launch inhibition (barrier C1) and command reordering could defeat abort or safe hold propagation (barrier C2).

5.5 Airspace control

Where airspace *must* be shared, failure to coordinate return corridors and defended volumes can expose friendly aircraft to weapon effects. Young (2005) notes that shared situational awareness must include defended assets, the C2 situation and defensive resources, as well as the track picture. Without this context, engagement decisions based on track kinematics and CID may still be unsafe.

These are primarily SoS resource conflict hazards arising from the competing uses of shared airspace (Redmond 2007). The mechanisms map primarily to the Initiating Event (IE) and barrier A of Figure 3, with further protection through procedural exclusion at barrier B, launch inhibition at barrier C1, and dynamic separation and intervention at barrier C2 (Section 4.2).

6 Hazard controls and allocation

For each causal mechanism, the analysis identified a function or constraint that could prevent, detect or contain it. In total, 44 hazard controls were identified: eight organisational or assurance controls, one aeronautical information service and 35 functional constraints. The controls were grouped into six categories:

a. Data/time integrity and provenance controls detect corruption, clock misalignment and sequencing errors, preserve source and processing history, and flag or restrict information whose integrity or timing cannot be established (Stouffer et al. 2023; Belhajjame et al. 2013).

b. Common reference frame and data controls align coordinate systems, datums, units and reference data versions, and verify transformations so that systems interpret positions and airspace boundaries consistently (Young 2005; Redmond 2007).

c. Airspace and FFT currency controls check receipt and currency of updates, flag stale, missing or conflicting information, and prevent its use in engagements where it cannot be relied upon to protect friendly forces (JCS 2010).

d. Confidence based inhibition controls prevent engagement when confidence in track quality, identification or supporting information falls below the required threshold. Stale, missing, ambiguous or conflicting evidence must therefore lead to restrictive action when that threshold is not met (CALL 2018; DoD 2025).

e. Independent safe-hold/abort paths provide a separate means of stopping an engagement while effective intervention remains possible. Their independence reduces the risk that one failure defeats both engagement control and protective intervention.

f. Cross enterprise integration and training align plans, procedures and responsibilities across participating units and authorities. Joint exercises establish a shared understanding of friendly activity, airspace use and engagement authority, and practise coordination under normal and degraded conditions (JCS 2010; JCS 2012).

Direct controls were allocated to event-tree barriers. The assurance, governance and support arrangements on which they depended were recorded separately as background conditions. Table 3 presents the Airspace Control subset. Controls were then allocated to managers, services, gateways or interfaces in the functional architecture (Table 4). A notional requirement, rationale and V&V objectives were developed to guide implementation and assessment (Table 1).[1]

7 Common cause failures

A count of hazard controls can overstate protection when several depend on the same service. Qualitative fault trees were used to identify dependencies at the functional level. The barrier A fault tree (Figure 4) identified several controls dependent on the aeronautical information service, HC-41. Loss or degradation of that service could therefore weaken several controls together.

The wider review also identified dependence on IFF, Link 16 and other shared services at the implementation level and that safety relies upon their availability (Leveson, Allen, and Storey 2002; GAO 1997; Altman 2025; Defence 1994). Their effects on the remaining barriers require further analysis.

8 Model evaluation

8.1 Development-case review

The four development cases of Section 2.4 could be mapped to the hazard taxonomy of Ericson (2015) and the functional threads (Table 2). Their fit to the resulting model provided a consistency check, not independent validation.

[1] This approach adapted the concept of mechanistic design abstraction described by Douglass (1999).

Table 1: Example allocation and derivation of hazard controls

Control *Solution agnostic*	SoS functional architecture *Allocated to*	Derived requirement *Solution aware (but not specific) derived requirement*
HC-09 | Cross-check against flight plan, mission, and sensor evidence	• CID Manager • Airspace Coordination Manager • Decision Support & Display Service	Candidate air tracks shall be cross-checked against mission plans, published routes, scheduled windows and available sensor corroboration before hostile status is asserted. *Rationale.* Context often distinguishes expected friendly behaviour from suspicious behaviour in the absence of positive identification (e.g., IFF or FFT). *Verification and Validation (V&V) objective.* Verify through scenario testing that route, procedure, and sensor context influence exclusion or inhibit logic.

8.2 Supplementary case evaluation

The 2003 Patriot shoot-down of a US Navy F/A-18 was assessed as a supplementary case not used to develop the model. The evaluation examined coverage by the existing taxonomy, functional threads, barrier structure and documented controls before identifying refinements. Although separate from the development cases, this event involved the same weapon system (Patriot) and campaign as the Tornado case.

The supplementary case exposed a gap, where the original H1 definition assumed that the friendly platform was not positively identified, and the event tree treated exclusion at barrier B as ending its exposure. A positively identified friendly aircraft could nevertheless be acquired during an engagement directed at another contact. H1 and the event tree were revised to carry this exposure through C1 and C2, and HC-44 was added to maintain separation from the missile's seeker acquisition and fly-out volumes. The existing hazard categories and functional threads remained applicable.

8.3 Limitations

The study has the following limitations:

a. *Case selection.* The small set of cases does not establish likelihood or demonstrate that all mechanisms have been identified.

b. *Source quality.* Evidence varied in scope, detail and authority. Reported findings, technical hypotheses and analytical assumptions were distinguished, but several aspects of the supplementary case remained unresolved.

c. *Retrospective interpretation.* Knowledge of the outcomes and qualitative mechanism coding introduced hindsight bias and analytical judgement into the assessment.

d. *Internal logic.* Case mapping checked the consistency of the hazard definition and event pathways, but did not validate every fault-tree gate, dependency or barrier-success condition.

e. *Control effectiveness.* The analysis identified and allocated controls but does not demonstrate their operational effectiveness or quantify their risk-reduction effects.

f. *Single engagement scenario.* The event tree considered only one engagement, leaving interactions among simultaneous engagements unassessed.

9 Discussion

The analysis showed that conventional event-tree and fault-tree techniques could support control allocation from functional information as part of a SoSHA. Including airspace control broadened the analysis to address conflicting uses of shared airspace, while dependency analysis showed how a common service failure could weaken several controls. The implications are discussed below:

(a) *Method.* Standard event and fault-tree techniques supported qualitative hazard tracing and control allocation using the available functional information. A detailed design and probability estimates were not required for this analysis.

(b) *Completeness.* Completeness, relative to the purpose of the analysis, is assessed through coverage of the identified causal mechanisms via traceability to controls.[2]

(c) *Analysis boundary.* The case review exposed the omission of airspace control from the original functional model. Adding it allowed the analysis to address conflicting uses of shared airspace.

(d) *Barrier dependencies.* Several controls depended on shared services. Operational availability of such systems must address the delivered service, including synchronisation, keys and usable data, rather than equipment uptime alone. Mission-entry and in-mission requirements should cover these dependencies and the communication paths needed to inhibit or terminate an engagement. Section 7 identifies where further dependency analysis is required.

(e) *Control allocation.* Background organisational controls enable active controls to operate effectively. Operational procedures were included in active controls when they directly constrained hazards.

(f) *Generalisability.* Functional thread tracing, event sequencing by epoch, control to barrier mapping and dependency analysis are transferable to other SoS applications.

9.1 Application to an Agile development

The JABMS design was already in development when the analysis began. Existing functions therefore had to be assessed against the proposed controls. Where a control was not implemented, it required new system requirements, allocation to operators, or both.

[2] This is an argument of sufficient rather than absolute completeness.

These allocations may change as later increments introduce greater automation.

Figure 5 shows how controls are checked against each increment and allocated to systems, operators or both. Control rationales and V&V objectives guide assessment of existing functions, and acceptance criteria for new functions. This includes off-the-shelf (OTS) behaviour where detailed requirements are unavailable (Table 1). This approach cannot however identify internal hazardous OTS dependencies.

The intended Agile delivery process records the required work in the backlog, with an owner and SoSHA acceptance criteria, and reviews the safety effects of each increment. Controls deferred beyond the relevant increment remain identified residual risks requiring the designated authority's acceptance.

10 Conclusion

The method combined functional threads with event-tree and fault-tree analysis to identify fratricide mechanisms and derive controls for allocation across the IAMD SoS. It provides a basis for assessing existing functions and specifying controls for successive capability increments. Their effectiveness remains to be demonstrated through implementation and verification. Advanced IFC capabilities such as remote fire and engagement handover are pacing requirements for fratricide prevention because they distribute the fire-control loop, system state and decision authority across constituent systems.

Further work should test the model against cases not used in its development and examine simultaneous engagements and cross-barrier dependencies. Controls should be tailored to individual IFC increments and traced through requirements to V&V evidence. Human and organisational controls require evaluation under realistic operating conditions.

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Table 2: Case study causal mechanisms and functional-thread mapping

Case study	Causal mechanisms	Threads	CM
Vincennes	Link 11 / IDS drift, and IFF-to-track mis-binding	T, C	IIF
	Civil and military users shared the same air and battlespace	A	IRC
	Track ID reuse leading to track data substitution	T, C	IRE
	Civil departure, mixed military airfield and engagement area were co-located	A	IPX
	Rapid shift from surface combat to air-defence decision-making under manoeuvre	C, E	RCN
	Climb state, altitude, IFF, and threat profile were interpreted inconsistently	C	INT
	Lack of a credible common aeronautical information service	C, A	INT
Black Hawk	AWACS/fighter/helicopter control and radio links misaligned	F, A	IIF
	Airspace not managed as a shared resource	A	IRC
	Terrain masking and high-speed visual identification	T, C	IPX
	Fighter sanitisation shifted to mixed helicopter/fighter operation	E, A	RCN
	Tasking, IFF, and visual cues were interpreted differently across the SoS	C	INT
Tornado	Autonomous battery operation and relay-only connectivity reduced access to COP	T, F	IIF
	Patriot battery had difficulty receiving Mode 1 (unencrypted) code	C	IIF
	Coalition recovery traffic and Patriot defensive arcs conflicted	E, A	IRC
	The descending recovery profile left little decision time	E	IPX
	Local autonomous self-defence operation occurred during aircraft return	E, A	RCN
	Generic anti-radiation threat criteria, permissive rules and non-IFF instructions	C, E	RCN
F/A-18 (s)	Adjacent-radar interference (possibly) generated a false ballistic-missile track	T	IPX
	Transition to ready triggered pre-enabled automatic engagement	E	RCN
	Friendly aircraft exposed to seeker acquisition during an engagement of false track	A, F, E	IRC
Gettysburg	Strike-group planning, watch handover, Link 16, IFF, and radar degraded	T, F, C, E	IIF
	Track picture degraded progressively due to re-tasking/loss of surveillance systems	T	IRE
	Returning aircraft, defensive sorties and expected hostile missiles in same airspace	A	IPX
	Carrier-recovery geometry and perceived threat axes reduced reaction time	E, A	IPX
	Abrupt shift to high-tempo engagement with watch and sensor-role changes	T, E	RCN
	Returning aircraft interpreted as hostile missiles due to engagement context change	C	RCN

Functional threads: T = tracking; F = friend force tracking; C = identification and classification; E = engagement; A = airspace control.

Mapping: Entries identify the threads directly affected by each mechanism. CM codes as per Figure 2. (s) denotes supplementary case.

Table 3: Airspace hazard controls (*Denotes dependency on HC-41)

Control ID	Description
HC-07	Positive receipt of mission and airspace updates*
HC-09	Cross-check against flight plan, mission, and sensor evidence*
HC-18	Procedural separation by altitude, heading, time, and manoeuvre*
HC-20	Airspace-currency checks
HC-26	Airspace, aviation & intelligence data integration
HC-28	Common spatial reference baseline*
HC-29	Spatial transformation and terrain-model conformance
HC-38	Protected-volume and return-corridor inhibit*
HC-41	Aeronautical information service (mission support)
HC-43	Airspace use deconfliction through static segregation measures
HC-44	Maintain separation from weapon seeker volume/fly-out path during missile engagement

Table 4: Functional architecture elements

Architectural	Definition	Example
Manager	A logical element that coordinates a defined enterprise function across constituent systems. It maintains the relevant state, applies decision rules and resolves or exposes conflicts, but need not be implemented as a single physical component.	A CID manager, for example, coordinates identification evidence and identity assessments.
Service	A bounded capability made available to other architectural elements through defined interfaces and service conditions. It provides information or performs a function without necessarily controlling how the receiving element uses the result.	A cooperative-IFF service, for example, performs interrogations and distributes validated replies.
Gateway	An element that enables information or control to pass between systems, networks or authority domains. It may translate, filter, validate, route or protect the exchange while preserving its meaning, context and any constraints.	An engagement-handoff gateway, for example, transfers engagement state and control information between participating systems.
Interface	A defined boundary through which architectural elements exchange information, commands or control. It specifies the content, format, timing, direction, authority and expected response associated with the exchange.	A weapon-control and abort interface, for example, conveys engagement commands and time-critical abort or ceasefire instructions to the weapon system.

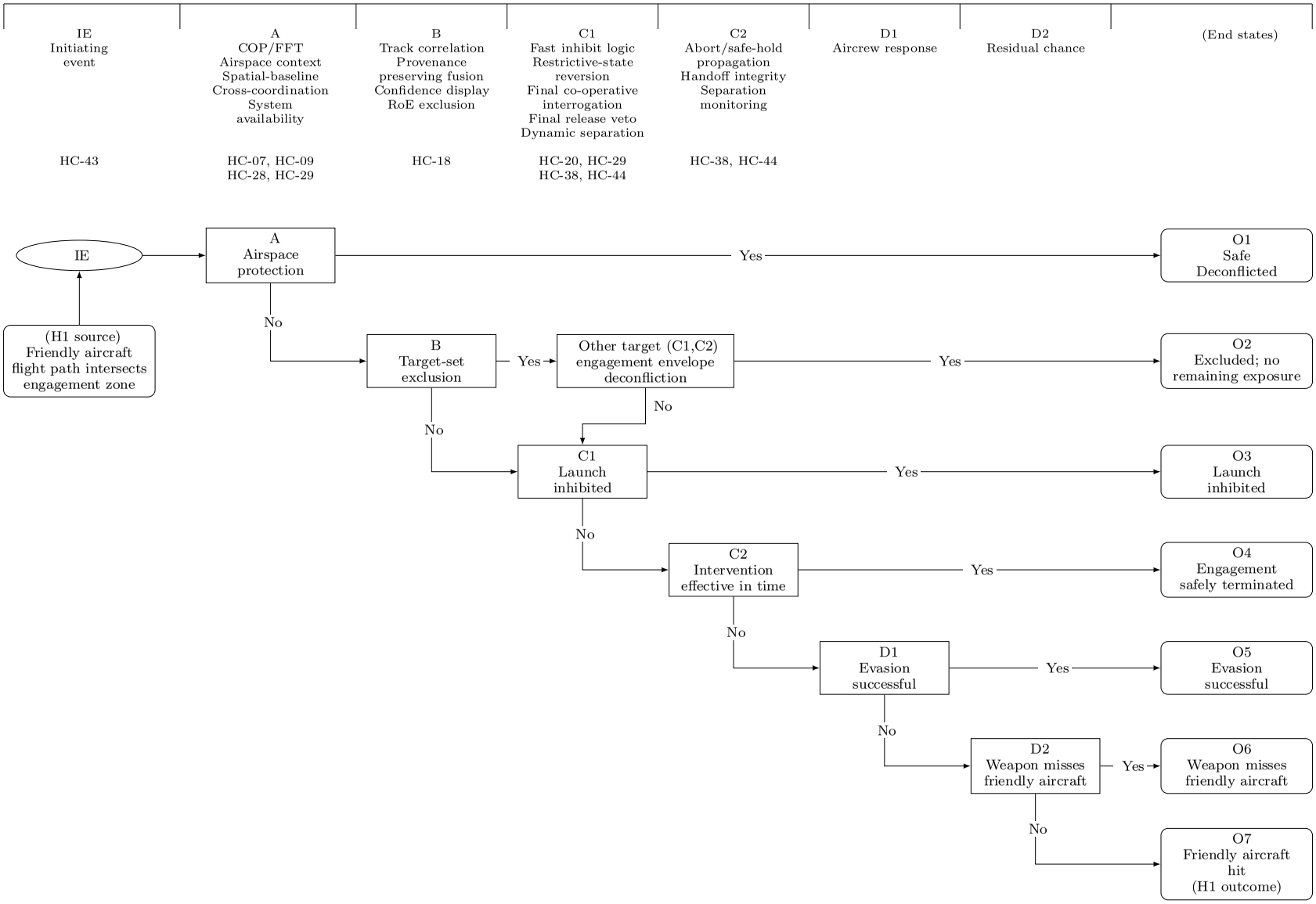
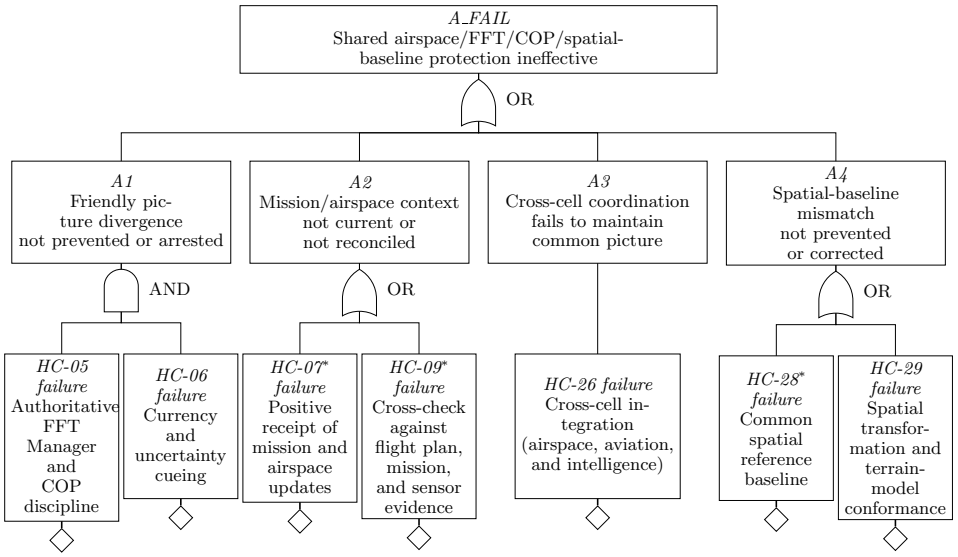


Figure 3: Fratricide event tree with airspace controls of Table 3 annotated



*Indicates CCF vulnerability to HC-41 service loss.

Diamonds denote undeveloped control-failure events.

The HC-26 leaf represents failure of operational information exchanges or coordination.

Figure 4: Barrier A fault tree

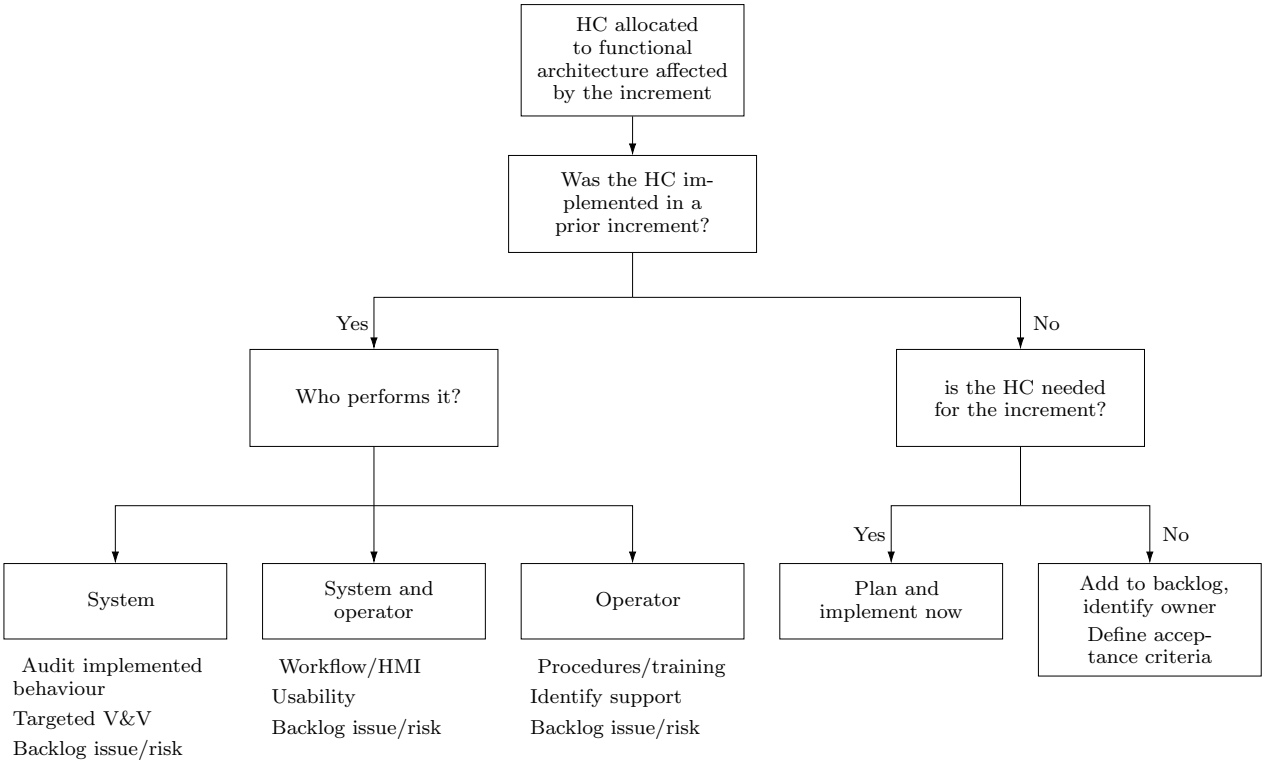


Figure 5: Agile increment decision process

Table 5: Key terms and definitions

Term	Definition
Combat identification (CID)[a]	The characterisation of detected objects sufficient to categorise them as friend, enemy, neutral or unknown, including their platform type and intent, to assess the threat posed and support an engage/no-engage decision (ADF 2009a).
Engagement chain	The linked sequence (thread) of functions, decisions and actions by which a potential target is detected, tracked, identified, classified, selected, authorised, engaged and subsequently subjected to kill assessment. In a SoS, the chain may cross multiple sensors, networks, command and control (C2) systems, decision authorities and weapon systems. See also kill chain (Becker et al. 2011).
Fratricide	The employment of friendly weapons and munitions with the intent to kill the enemy or destroy their equipment or facilities, which results in unforeseen and unintentional death or injury to friendly personnel (Headquarters, Department of the Army 1996). The decision to engage may appear locally rational in light of all the information available to the decision maker at the time.
Fratricide hazard (SoS)	The SoS fratricide hazard defined in Section 4 (H1).
Friend Force Tracking (FFT)	The process of fixing, observing, and reporting the location and movement of friendly forces, to support situational understanding, command and control, combat identification and the prevention of fratricide (Staff 2010).
Functional thread (system)	A functional or causal path through the constituent systems (and their hardware, software and human elements) that make up the SoS (Whitford 2015). Similar in concept but not identical to computer software processing threads.
Joint Engagement Zone (JEZ)	In air defence, that airspace of defined dimensions within which multiple air defence systems (surface-to-air missiles and aircraft) are simultaneously employed to engage air threats (JCS 2010).
IAMD SoS (ADF)	The ADF's networked combination of independently managed sensors, C2 systems, communications networks and weapon system platforms that performs joint engagement chains across domains. As part of this enterprise the JABMS acts to integrate information and coordinate constituent systems in order to generate a Common Operational Picture (COP) and enable the most appropriate effector to act (Australian DoD 2026).
Integrated Fire Control (IFC)	IFC refers to the collaboration of multiple distributed constituent systems making up the IAMD SoS in the detection, identification and engagement of enemy targets (Australian DoD 2022). This collaboration in turn delivers SoS level capabilities such as remote threat cueing and the ability to pass control of a missile engagement from a launch system to a remote unit (Young 2005).
Restrictive state reversion	If required engagement permissives become stale, contradictory or unavailable, the engagement state of a target automatically reverts to a safer state, such as hold, unknown or engagement prohibited.
Seeker acquisition volume	The volume of airspace within which a missile seeker may acquire an object for tracking at a specified time, determined by the missile's position and orientation, the seeker's field of view and its acquisition limits. For safety analysis, the predicted volume accounts for uncertainty in these factors.
Sensor net	An integrated system of sensor strings, relays, and monitoring sites established to provide sensor surveillance over all or part of the area of operation (USMC 2022).
Track	The function of associating successive sensor detections and other observations with an object, and estimating and maintaining that object's state over time within the surveillance area (Koch 2006; JCS 2012). Modern air defence systems also use tracking data (e.g., self reporting or kinematic) as an input to identification and classification functions to decide what is a threat and whether to engage it.
Tracklet	A tracklet is a short sequence of sensor observations associated with the same object, forming a segment of its track. In sensor netting, tracklets allow observations from different sensors and times to be correlated and fused into a common track, helping maintain track continuity across sensor coverage areas (Vananti et al. 2017).

[a] CID and FFT are treated as two separate functions because fratricide prevention via deconfliction depends on both threat and friendly force position awareness.